Basic and Applied Thermodynamics
– P.K. Nag.

Macroscopic vs Microscopic Viewpoint:

* In the microscopic approach, a certain quantity of matter is considered, without the events occurring at the molecular level being taken into account.

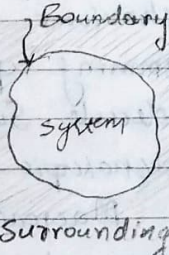
* From the microscopic point of view, matter is composed of myriads of molecules. If it is a gas, each molecule at a given instant has a certain position, velocity, and energy and for each molecule these change very frequently as a result of collisions. The behaviour of the gas is described by summing up the behaviour of each molecule. Such a study is made in microscopic or statistical thermodynamics.

Macroscopic thermodynamics is only concerned with the effects of the action of many molecules and the effects can be perceived by human senses.

Ashcroft & Mermin
Solid State Physics

Thermodynamic system and control volume:

A thermodynamic system is defined as a quantity of matter or a region in space upon which attention is concentrated in the analysis of a problem. Everything external to the system is called the surroundings or the environment. The system is separated from the surroundings by the system boundary. The boundary may be either fixed or moving. A system and its surroundings together comprise a universe.



Three classes of systems:

a) closed system:

A system of fixed mass. There is no mass transfer across the system boundary. There may be energy transfer into or out of the system.

b) Open system:

A system in which matter crosses the boundary of the system. There may be energy transfer also.

c) Isolated system:

A system in which there is no interaction between the system and the surroundings. It is of fixed mass and energy, and therein no mass or energy transfer across the system boundary.

Control volume:

For thermodynamic analysis of an open system, attention is focussed on a certain volume in space surrounding the system, known as the control volume, bounded by a surface called the control surface. Matter as well as energy crosses the control surface.

A closed system is a system closed to matter flow, though its volume can change against a flexible boundary. When

there is matter flow, then the system is considered to be a volume of fixed identity, the control volume. There is thus no difference between an open system and a control volume.

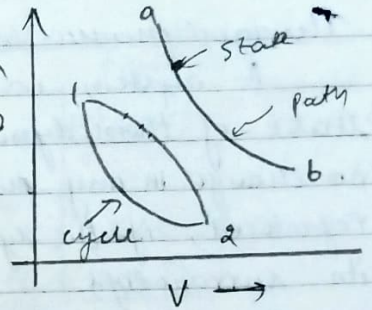
Thermodynamic properties, processes and cycles:

Properties of the system:

Volume, temperature, pressure etc. These are all macroscopic in nature. When all the properties of a system have definite values, the system is said to exist at a definite state. Properties are the coordinates to describe the state of a system. These are the state variables.

* Change of state: Succession of states passed through during a change of state is called the path of the change of state. When the path is completely specified, the change of state is called a process. If the final state is same as initial - cycle.

* Intensive properties:
Independent of mass in the system
eg: pressure, temperature
* Extensive properties:
Related to mass.
eg: volume, energy, density.
* Specific extensive properties: (per unit mass)
eg: specific volume, specific energy, density etc.



Homogeneous and Heterogeneous systems:

Homogeneous throughout in chemical composition and physical structure is called a phase. (solid, liquid, gas).
A system consisting of a single phase is called a homogeneous system, while a system consisting more than one phase is known as a heterogeneous system.
Eg: Ice floating on water is heterogeneous

Thermodynamic equilibrium:

A system is said to be in a state of thermodynamic equilibrium when no change in any macroscopic property is registered, if the system is isolated from its surroundings.

An isolated system always reaches in course of time a state of thermodynamic equilibrium and can never depart from it spontaneously.

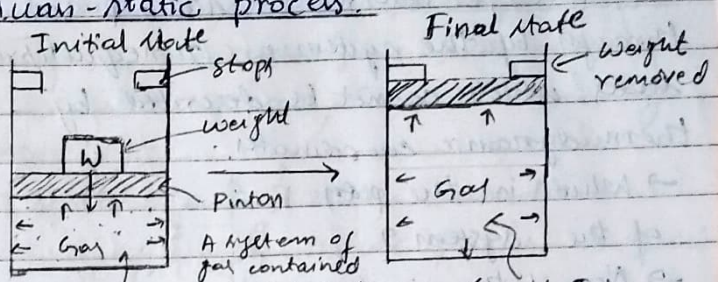
A system will be in a state of thermodynamic equilibrium, if the conditions for the following three types of equilibriums are satisfied.

(a) Mechanical equilibrium
(b) Chemical equilibrium
(c) Thermal equilibrium.

→ Therefore, there can be no spontaneous change in any macroscopic property if the system exists in an equilibrium state.

Thermodynamics studies mainly the properties of physical systems that are found in equilibrium states.

Quasi-static process:



Transition between two equilibrium states by an unbalanced force

→ The system initially in in an equilibrium state with properties P_1, V_1, T_1. The weight on the piston just balances the upward force exerted by the gas.

→ If the weight is removed, there will be an unbalanced force between the system and the surroundings, and under gas pressure, the piston will move up till it hits the stops.

→ The system again comes to an equilibrium state with new properties P_2, V_2, T_2.

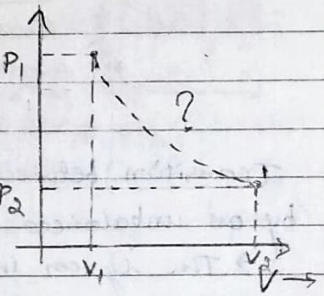
→ But the intermediate states passed through by the system are nonequilibrium states which cannot be described by thermodynamic coordinates.

→ What is the path of the system?

→ Now if the single weight on the piston is made up of many very ~~well~~ small pieces of weight and those weights are removed one by one very slowly from the top of the piston, at any instant of the upward travel of the piston, if the ~~so~~ gas system is isolated, the departure of the state of the system from the thermodynamic equilibrium state will be infinitesimally small.

→ So every state passed through by the system will be an equilibrium state.

→ Such a process, which is but a locus of all the equilibrium points ~~pas~~ passed through by the system, is known as a quasi-static process, 'quasi' meaning 'almost'.



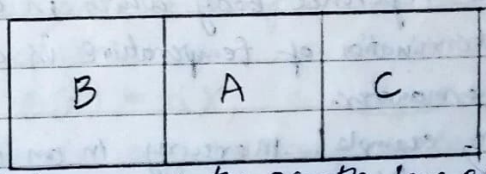
Transition between two equilibrium states

→ Infinite slowness in the characteristic feature of a quasi-static process. A quasi-static process is thus a succession of equilibrium states.

→ A quasi-static process is also called a reversible process.

Zeroth law of thermodynamics:

"When a body A is in thermal equilibrium with a body B and also separately with a body C, then B and C will be in thermal equilibrium with each other."



This is known as the zeroth law of thermodynamics. It is the basis of temperature measurement.

Temperature is associated with the ability to distinguish hot from cold. When two bodies at different temperature are brought into contact, after some time they attain a common temperature and are then said to exist in thermal equilibrium.

Thermometer:

"To obtain a quantitative measure of temperature, a reference body is used and a certain physical characteristic of that body which changes with temperature is selected. The changes in the selected characteristic may be taken as an indication of change in temperature. The selected characteristic is called thermometric property and the reference body which is used in the determination of temperature is called the thermometer.

Common example: Mercury in an evacuated capillary tube. Expansion of mercury is the thermometric property.

Five different kinds of thermometers:

Thermometer	Thermometric property
1. Constant volume gas thermometer	Pressure (p)
2. Constant pressure gas thermometer	Volume (V)
3. Electrical resistance thermometer	Resistance (R)
4. Thermocouple	Thermal emf (ε)
5. Mercury-in-glass thermometer	Length (L)

Temperature measurement - Reference points

If X is the thermometric property, let us arbitrarily choose for the temperature common to the thermometer used and to all systems in thermal equilibrium with it the following linear function of X:

$$\theta(X) = aX,$$

where a is the proportionality constant.

If X_1 corresponds to $\theta(X_1)$, then

$$\theta(X_1) = aX_1$$

and $$a = \frac{\theta(X_1)}{X_1}$$

also then X_2 will correspond to temperat.

$$\theta(X_2) = \frac{\theta(X_1)}{X_1} X_2 .$$

Method used before 1954:

$$\frac{\theta(X_1)}{\theta(X)} = \frac{X_1}{X}$$ Two Standards

and $$\frac{\theta(X_2)}{\theta(X)} = \frac{X_2}{X}$$

From these two.

$$\frac{\theta(X_1) - \theta(X_2)}{\theta(X)} = \frac{X_1 - X_2}{X}$$

or $$\theta(X) = \frac{\theta(X_1) - \theta(X_2)}{X_1 - X_2} \cdot X$$

$\theta(X)$ can be calculated from the measurement of X, X_1 and X_2 (thermometric properties) and arbitrarily assigning values for $\theta(X_1)$ and $\theta(X_2)$.

An easily reproducible state of an arbitrarily chosen standard system is called a fixed point. Before 1954 there were two fixed points: (a) the ice point (b) the steampoint at 1 atmospheric pressure. These were chosen to be 0 and 100°C.

The use of two fixed points was found unsatisfactory and later abandoned because of;

(a) the difficulty of achieving equilibrium between pure ice and air-saturated water (since when ice melts, it surrounds itself only with pure water and prevents intimate contact with air-saturated water), and (b) extreme sensitiveness of the steam point to the change in pressure.

Method used after 1954:

Since 1954 only one fixed point has been in use, viz, "the triple point of water", the state at which ice, liquid water and water vapour coexist in equilibrium. The temperature at which this state exists is arbitrarily assigned the value of 273.16 kelvin or 273.16 K, (equal to 0.01°C). If, (at 0.006036 59 atm (1atm = 101325 Pa) / (1 Pa = 1 Nm⁻²))

θ_t = the triple point of water

X_t = value of thermometric property when the body, whose temperature θ is to be measured, is placed in contact with water at its triple point,

it follows that,

$$\theta_t = aX_t$$

$$a = \frac{\theta_t}{X_t} = \frac{273.16}{X_t}$$

Therefore for unknown temperature,

$$\theta = aX$$

$$= \frac{273.16}{X_t} X = 273.16 \frac{X}{X_t}$$

The temperature of the triple point of water, which is an easily reproducible state, is now the standard fixed point of thermometry.

Comparison of thermometers:

Applying the above principle to the five thermometers listed, the temperatures are,

a) Constant volume gas thermometer

$$\theta(P) = 273.16 \frac{P}{P_t}$$

b) Constant pressure gas thermometer

$$\theta(V) = 273.16 \frac{V}{V_t}$$

c) Electric resistance thermometer

$$\theta(R) = 273.16 \frac{R}{R_t}$$

d) Thermocouple

$$\theta(\mathcal{E}) = 273.16 \frac{\mathcal{E}}{\mathcal{E}_t}$$

e) Liquid in glass thermometer.

$$\theta(L) = 273.16 \frac{L}{L_t}$$

Ideal gas:

R = Characteristic gas constant.

$P\bar{v} = \bar{R}T$, $\bar{R}$ = universal gas constant $= 8.3143$ J/mol K

$\bar{v}$ = molar specific volume in $m^3/g\,mol$

$PV = n\bar{R}T$, $PV = mRT$

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$ | R | n = no. of moles, m = mass of

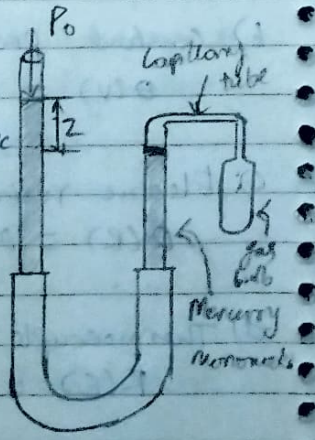
At very low pressure or density all gases and vapours approach ideal gas behaviour.

Gas thermometer:

The pressure in the bulb is used as a thermometric property and is given by

$p = p_0 + \rho_M Z g$

where p_0 is the atmospheric pressure, ρ_M is the density of mercury.



When the bulb is brought in contact with the system whose temperature is to be measured, the bulb, in course of time comes in thermal equilibrium with the system. The gas in the bulb expands, on being heated, pushing the mercury downward. The flexible limb of the manometer is then adjusted so that the mercury again touches the tip L. The difference in mercury level Z is recorded and the pressure p of the gas in the bulb is estimated. Since the volume of the trapped gas is constant, from the ideal equation,

$$\Delta T = \frac{V}{mR}\Delta p$$

i.e, the temperature increase is proportional to the pressure increase. As a reference if the triple point is used

$\Delta T \propto \Delta P$

for an ideal gas at constant volume,

$T \propto P$

$T_t \propto P_t$ ← triple point

gives, $\frac{T}{T_t} = \frac{P}{P_t}$

or $T = \frac{T_t}{P_t} P = \frac{273.16}{P_t} P$

If operated with ordinary gas,

$$T = 273.16 \lim_{P_t \to 0} \left[\frac{P}{P_t}\right]$$

In a constant pressure gas thermometer, the mercury levels have to be adjusted to keep Z constant, and the volume of gas V, which would vary with the temperature of the system, becomes the thermometric property.

$$\Delta T = \frac{P}{R} \Delta V$$

i.e., the temperature increase is proportional to the observed volume increase.

The constant volume gas thermometer is, however, mostly in use, since it is simpler in construction and easier to operate.



Celsius temperature scale:

It employs a degree of the same magnitude as that of the ideal gas scale but its zero point is shifted.

For triple point of water

$$t = 0.01\ ^\circ C$$
$$= 273.16 - 273.15$$
$$t = T - 273.15$$

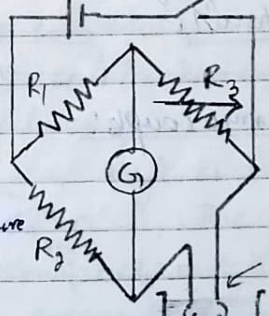
(t in °C scale, T in Kelvin scale)

For steam point at 1 atm,

$$t_s = T_s - 273.15$$
$$= 373.15 - 273.15 = 100^\circ C.$$

Electrical resistance thermometer:

In the resistance thermometer the change in resistance of a metal wire due to its change in temperature is the thermometric property. The wire



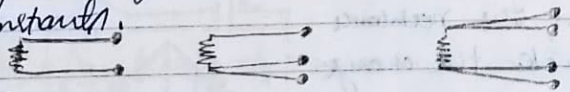
$$\frac{R_1}{R_2} = \frac{R_3}{R} \quad \text{or} \quad R = \frac{R_2}{R_1} R_3$$

frequently platinum, may be incorporated in a wheatstone bridge circuit. The platinum resistance thermometer measures temperature to a high degree of accuracy and sensitivity, which makes it suitable as a standard for the calibration of other thermometers

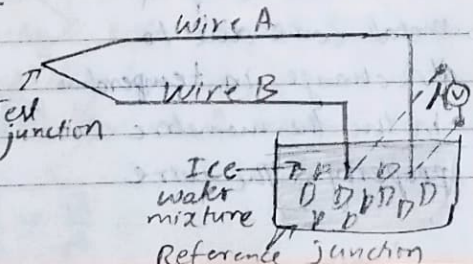
In a restricted range, the following quadratic equation is often used,

$$R = R_0(1 + At + Bt^2)$$

where R_0 is the resistance of the platinum wire when it is surrounded by melting ice and A and B are constants.



Thermocouple:



A thermocouple is made up of joining two wires A and B made of dissimilar metals. Due to Seebeck effect, a net emf is generated in the circuit which depends on the difference in temperature between the hot and cold junction and is, therefore a thermometric property of the circuit. This emf can be measured by a microvoltmeter to a high degree of accuracy. The choice of metals depends largely on the temperature range to be investigated, and copper-constantan (copper + Nickel) (55% 45%) (Nickel + Chromium) (90 10) chromel-alumel (Nickel + Al + Mn + Si) (95% 2% 2% 1%) platinum-platinum + Rhodium are typical combinations in use.

A thermocouple is calibrated by measuring the thermal emf at various known temperatures, the reference junction being kept at 0°C. The results of such measurements on most thermocouples can usually be represented by a cubic

equation of the form:

$$\mathcal{E} = a + bt + ct^2 + dt^3$$

where $\mathcal{E}$ is the thermal emf and the Constants a, b, c, d are different for each thermocouple

Liquid in glass thermometer:

Mercury or alcohol → −97.78°C (freezing)
64.65°C (boiling)
356.72°C (boiling)

A liquid in glass thermometer measures temperature based on the thermal expansion of mercury or spirit alcohol in a glass container.

	Alcohol	Mercury
Melting point	−97.78°C	−38.86°C
Boiling point	64.65°C	356.72°C

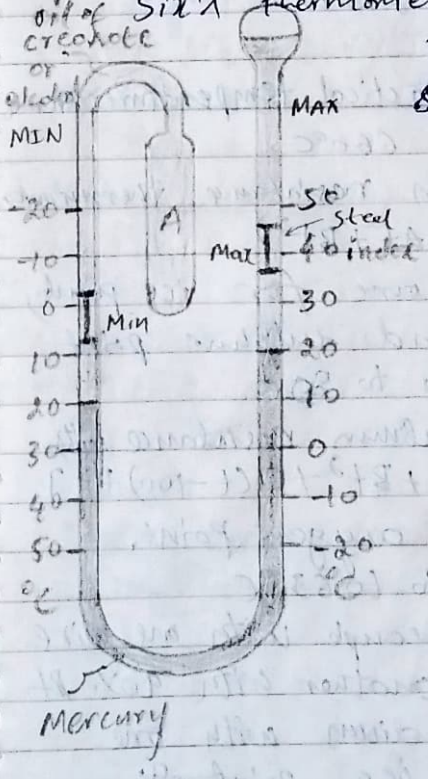


Because mercury has low thermal capacity, high heat conductivity, inertness in relation to a glass capillary tube and a high boiling point, it is an ideal thermometric liquid except for its relatively high melting point.

Maximum - minimum thermometer:

Six's thermometer:

Invented by James Six in 1782



Scales of Temperature:

$F = \frac{9}{5} C + 32$

$1K = \frac{1}{273.16}$ of triple point

$T_{°C} = T_K - 273.15$

International practical temperature scale

a) From 0 to 660°C

Platinum resistance thermometer

$R = R_0 (1 + At + Bt^2)$

A, B and R_0 are for ice point, steam point and sulphur point

b) From −190 to 0°C

Same platinum resistance with

$R = R_0 [1 + At + Bt^2 + C(t-100)t^3]$

constants from oxygen point.

c) From 660 to 1063 °C.

Thermocouple with one wire platinum and another with 90% Pt and 10% rhodium with one junction at ice point 0°C.

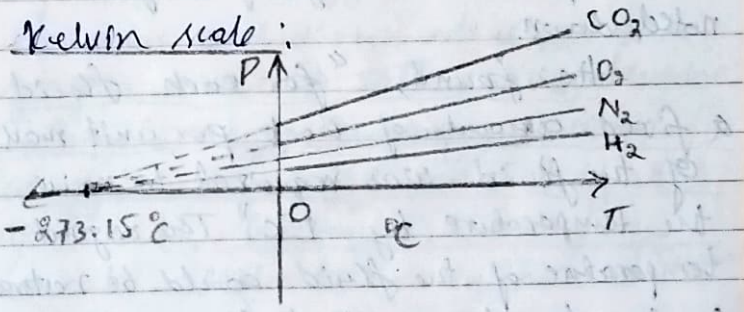
$$\mathcal{E} = a + bt + ct^2$$

a, b, c from antimony point, silver point and gold point. Dia of wire between 0.35 and 0.65 mm.

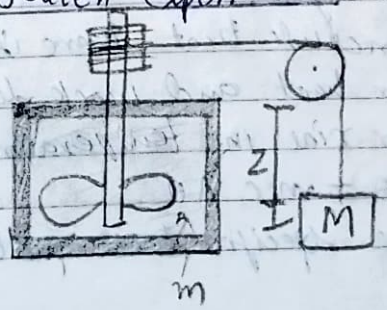
d) Higher than gold point:

By black body radiation with Planck's law of thermal radiation

Kelvin scale:



Joule's experiment:



$Mgz = \Delta E$

$= mc\Delta T$.

James Prescott Joule

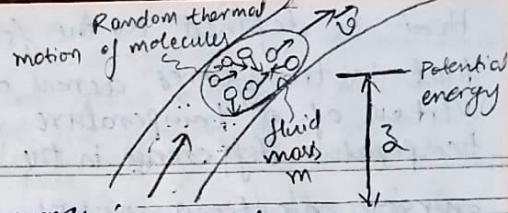
James P Joule carried out his famous experiment. He placed known amount of water, oil, mercury in an insulated container and agitated the fluid with a rotating stirrer. The amount of work done on the fluid by the stirrer were accurately measured (mgz) and the temperature changes of the fluid were noted down.

He found, "for each fluid a fixed amount of work per unit mass of the fluid was required to raise the temperature by $1^{\circ}C$." The original temperature of the fluid could be restored by the transfer of heat through contact with a cooler object.

You can conclude that there is a relation between heat and work done on the liquid for rise in temperature.

$$Mgz = \Delta Q = mc\,\Delta\theta$$

where, c is the specific heat capacity of the fluid.



Internal energy: stored in a

E = total energy of the system

$= E_{macroscopic} + E_{microscopic}$

$= [E_k + E_p] + U$

where, $E_{macro} = E_k + E_p$

$E_{micro} = U$

The microscopic energy mode refers to the energy stored in the molecular and atomic structure of the system, which is called the molecular internal energy or simply internal energy, customarily denoted by the symbol U. Matter is composed of molecules. Molecules are in random thermal motion (for a gas) with an average velocity $\bar{v}$, constantly colliding with one another and with the walls. Due to a collision, the molecules may be subjected to rotation as well as vibration. They can have translational kinetic energy, rotational kinetic energy, vibrational

Heat is defined as the form of energy that is transferred across a boundary by virtue of a temperature difference. The temperature difference is the 'potential' or force

energy, electronic energy, chemical energy and nuclear energy. If ε represents the energy of one molecule then,

$$\varepsilon = \varepsilon_{trans} + \varepsilon_{rot} + \varepsilon_{vib} + \varepsilon_{chem} + \varepsilon_{electronic} + \varepsilon_{nuclear}$$

If N is the total number of molecules in the system, then total internal energy

$$U = N\varepsilon$$

In an ideal gas there are no intermolecular forces of attraction and repulsion, and the internal energy depends only on temperature. Thus

$$U = f(T) \text{ only.}$$

In absence of other forms such as surface (tension) energy, electrical energy, magnetic energy,

$$E = \underbrace{E_K + E_P}_{\text{macro}} + \underbrace{U}_{\text{micro}}$$

If $E_K = 0$, $E_P = 0$, $\therefore E = U$

$$\therefore Q = \Delta U + W$$

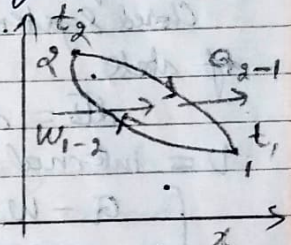
U is an extensive property of the system (depends on size of the system).

and heat transfer in the fluid.

Heat flow into a system is taken to be positive, and heat flow out of a system is taken as negative. The symbol Q is used for heat transfer i.e. quantity of heat transferred within a certain time.

First law of Thermodynamics:

Converting ~~in~~ a system executing cyclic a thermodynamic cycle, in which consists of a definite amount of work input W_{1-2} to the system followed by a transfer of an amount of heat Q_{2-1} from the system.



It has been found that W_{1-2} is always proportional to the heat Q_{2-1} and the constant of proportionality is called the Joule's equivalent or the mechanical equivalent of heat.

$$(\Sigma W) \; W_{1-2} \propto Q_{2-1}$$

$$(\Sigma W)_{cycle} = J(\Sigma Q)_{cycle}$$

where J is the Joule's equivalent.

Also expressed as,

$$\oint dW = J \oint dQ, \quad J = 1 \text{ Nm/J}$$

$$\therefore \oint dW = \oint dQ$$

"Energy is neither created nor destroyed".

This is the first law for a closed system undergoing a cycle.

Corollary 1:

Closed system undergoing a change of state.

$$dE = dQ - dW$$

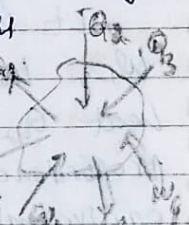
U = internal energy, E = total energy = internal energy + KE + P.E + electrical energy + magnetic + chemical

$Q - W = \Delta E$

or $Q = \Delta E + W$

- If there are more energy transfer quantities involved in the process

$$(Q_2 + Q_3 - Q_1) = \Delta E + (W_2 + W_3 - W_1 - W_4)$$



Energy is thus conserved in the operation. The first law is a particular formulation of the principle of the conservation of energy. Equation $\Delta E = Q - W$ may also be considered as the definition of energy. This definition does not give 'an' absolute value of energy E, but only the change of energy ΔE for the process. It can, however, be shown that the energy has a definite value at every state of a system and is, therefore, a property of the system.

Corollary 2:

For the isolated system, heat and work both interactions are absent ($dQ = 0$, $dW = 0$) and U is a constant. Energy can neither be created nor be destroyed but it can be converted from one form to other.

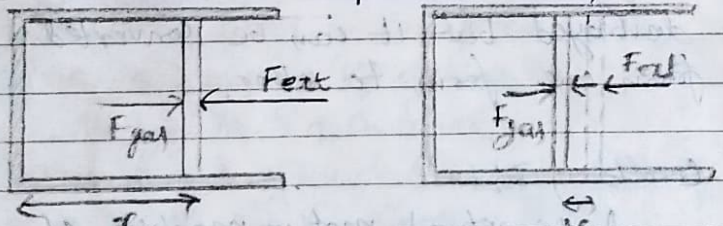
Corollary 3:

A perpetual motion machine of first kind (PMM 1) is impossible as it violates the 1st law of thermodynamics. PMM 1 is a hypothetical machine which can produce useful work without any source or which can produce more energy than consumed. First law says that energy can be neither created nor destroyed. A PMM 1 violates the

have by creating energy. PMM1 does not and can never exist.

Eg: A generator (electric) producing electricity which inturn used to run the generator itself.

Work done in ideal gas process: (PdV work or displacement work)



Work done by a force F on the system as it moves the object from position s_i to position s_f is,

$$W = \int_{s_i}^{s_f} \vec{F} \cdot d\vec{s}$$

If displacement in along the direction of the force then,

$$W = \int_{s_i}^{s_f} F\, ds \quad [\cos\theta = 1].$$

Let us calculate "work done on the" gas by an external force F_{ext}. This work it has to do against force exerted by the gas on the piston. Hence,

$$F_{ext} = -F_{gas} = -pA$$

Suppose the piston moves a small distance dx to the left, then the external force does an amount of work of

$$dW = F_{ext}\, dx = -pA\, dx$$

If dx is $-ve$, dw is $+ve$
If dx is $+ve$, dw is $-ve$

Change in volume of the gas is,

$$dV = A\, dx$$

Consequently work done is,

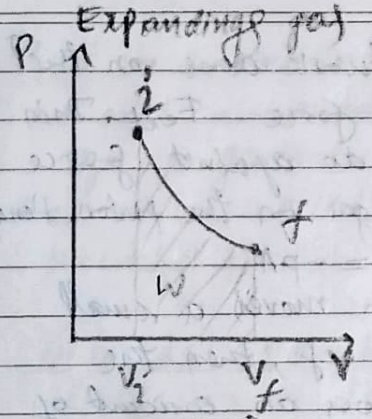
$$dw = -p\, dV$$

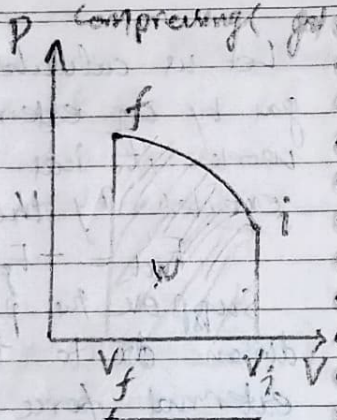
If in the process the volume changes from V_i to V_f the total work done by F_{ext} is the integral,

$$W = -\int_{V_i}^{V_f} p\, dV$$

To evaluate we need to know

$$P \equiv P(V)$$



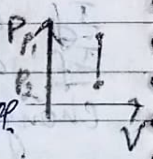


Work done in:

Isochoric process:

$W_{if} = -\int P\,dV = 0$

Since volume doesn't change.



Isobaric process:

Pressure remains constant and volume changes from V_i to V_f.

$W_{if} = -\int_{V_i} P\,dV = -P\int_{V_i} dV$

$= -P(V_f - V_i) = -P\Delta V$

If ΔV is positive (gas expands) → then W is negative.

If ΔV is negative (gas compresses) → then W is positive.

Isothermal process:

For ideal gas, $PV = n\bar{R}T$, $P = \frac{n\bar{R}T}{V}$

$\therefore W = -\int_{V_i}^{V_f} P\,dV$

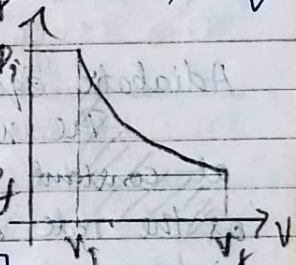
$= -n\bar{R}T\int_{V_i}^{V_f} \frac{dV}{V}$

$= -n\bar{R}T\left[\ln(V)\right]_{V_i}^{V_f}$

$= -n\bar{R}T\left[\ln(V_f) - \ln(V_i)\right]$

$= -n\bar{R}T\ln\left(\frac{V_f}{V_i}\right)$ $\left[\because PV = nRT,\ P_iV_i = P_fV_f\right]$

or $= -P_iV_i\ln\left(\frac{V_f}{V_i}\right) = -P_fV_f\ln\left(\frac{V_f}{V_i}\right)$



Adiabatic expansion of gas:

The process in which no heat enters the system or leave the system to the surroundings in an adiabatic process. The corresponding changes in

volume and pressure are said to be adiabatic changes. Thus in first law of thermodynamics, we can say that for a adiabatic process,

$$dQ = dU + dW = 0$$

Adiabatic equation of state:

The specific heat of a substance at constant volume C_V is defined as the rate of change of specific internal energy with respect to temperature when the volume is held constant, i.e,

$$C_V = \left(\frac{\partial u}{\partial T}\right)_V \quad \text{where } u = \frac{U}{m}$$

For a constant volume process.

$$Q_V = (\Delta U)_V = \int_{T_1}^{T_2} C_V \, dT$$

$$\therefore dU = m C_V dT \quad dW = P dV \quad \Big\} \Rightarrow dQ = dU + dW = 0 \text{ (adiabatic)}$$

Thus $dQ = m C_V dT + P dV = 0. \longrightarrow (1)$

So we arrive at,

$$m C_V dT = -P dV \longrightarrow (1)$$

Ideal gas equation $\longrightarrow$ (2)

$$PV = mRT \quad \text{where } R \text{ is the characteristic gas constant}$$

where $R = \frac{R_0}{M}$ in J/kg K and R_0 is universal gas constant.

Differentiating eq (2),

$$P dV + V dP = m R dT$$

$$\Rightarrow dT = \frac{P}{mR} dV + \frac{V}{mR} dP \longrightarrow (3)$$

Substituting (3) into (1) we get,

$$m C_V \left(\frac{P}{mR} dV + \frac{V}{mR} dP\right) + P dV = 0$$

$$C_V (P dV + V dP) + R P dV = 0$$

$$(C_V + R) P dV + C_V V dP = 0 \longrightarrow (4)$$

But $C_P - C_V = R$

or $C_P = C_V + R$

where $C_P = \left(\frac{\partial h}{\partial T}\right)_P$ is specific heat

at constant pressure. Here.

$H = U + pV$ in total enthalpy and $h = \frac{H}{m}$ (J/kg). is the specific enthalpy.

Hence eqn(4) becomes,

$$C_p P dV + C_v V dP = 0$$

Dividing throughout by $C_p PV$ we get

$$\frac{dV}{V} + \frac{C_v}{C_p}\frac{dP}{P} = 0$$

or $$\frac{C_p}{C_v}\frac{dV}{V} + \frac{dP}{P} = 0$$

or $$\gamma \frac{dV}{V} + \frac{dP}{P} = 0 \quad \text{where } \gamma = \frac{C_p}{C_v}$$

Hence after integrating,

$$\gamma \ln V + \ln P = \text{constant}$$

$$\ln V^\gamma + \ln P = \text{constant}$$

$$\ln(V^\gamma P) = \text{constant}$$

or $$PV^\gamma = \text{constant}$$

Variants:

1) $PV = mRT \Rightarrow P\left(\frac{mRT}{P}\right)^\gamma = \text{constant}$

$\therefore V = \frac{mRT}{P}$

$\Rightarrow (mR)^\gamma T^\gamma P^{1-\gamma} = \text{constant}$

2) $PV = mRT$

$P = \frac{mRT}{V}$

$\therefore \frac{mRT}{V} V^\gamma = mRTV^{\gamma-1} = \text{constant}$

Work done in adiabatic expansion of gases:

For adiabatic expansion,

$PV^\gamma = K$ (where K is a constant)

$\therefore P = \frac{K}{V^\gamma}$

$\therefore dW = PdV = \frac{K}{V^\gamma} dV$

By integrating it, total work done is,

$$W = \int_{V_i}^{V_f} \frac{K}{V^\gamma} dV = K\left[\frac{V^{-\gamma+1}}{-\gamma+1}\right]_{V_i}^{V_f}$$

$$= \frac{K}{1-\gamma}\left[V_f^{1-\gamma} - V_i^{1-\gamma}\right]$$

Also, $K = PV^\gamma = P_i V_i^\gamma = P_f V_f^\gamma$

$$\therefore W = \frac{1}{1-\gamma}\left[K V_f^{1-\gamma} - K V_i^{1-\gamma}\right]$$

$$= \frac{1}{1-\gamma}\left[P_f V_f^\gamma V_f^{1-\gamma} - P_i V_i^\gamma V_i^{1-\gamma}\right]$$

$$W = \frac{1}{1-\gamma}\left[P_f V_f - P_i V_i\right]$$

This is the work done by the system during an adiabatic process.

$1\,Pa = 1\,N m^{-2}$

$1\,bar = 10^5\,Pa = 100\,kPa = 0.1\,MPa$

$1\,atm = 101.325\,kPa = 1.01325\,bar$

$1\,atm = 760\,mm$ of Hg.

Numericals:

1. The pressure of a gas in a pipeline is measured with a mercury manometer having one limb open to the atmosphere. If the difference in the height of mercury in the two limbs is 562 mm calculate the gas pressure. The barometer reads 761 mm Hg. The acceleration due to gravity is 9.79 m/s^2 and the density of mercury is 13640 kg/m^3.

Since one limb of the manometer is open (P_{atm}) we have,

$$P = P_0 + \rho g Z$$

At the plane AB (lower plane) we have,

$P = P_0 + \rho g z$ and $P_0 = \rho g z_0$

where z_0 is the barometric height ρ is density of mercury and P_0 is atmospheric pressure. Therefore,

$$P = \rho g z_0 + \rho g z = \rho g (z_0 + z)$$

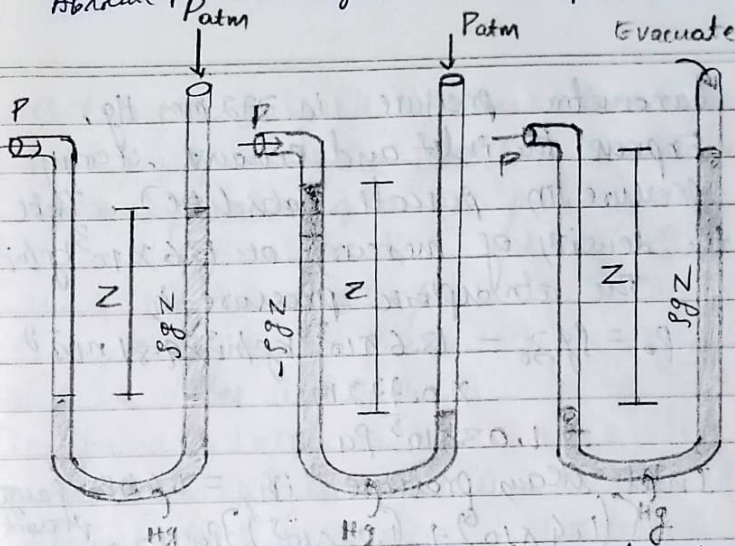
$$= 13640\,kgm^{-3} \times 9.79\,ms^{-2} \times (0.562 + 0.761)\,m$$

$$= 177 \times 10^3\,N m^{-2} = 177\,kPa = 1.77\,bar$$

$$= 1.746\,atm$$

The pressure relative to atmosphere is called gauge pressure. Relative to vaccum it's absolute pressure.

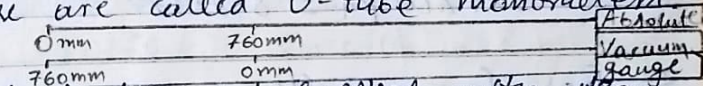
Absolute Pressure = Gauge pressure + Atmospheric pressure.



Open U-tube indicating gauge pressure	Open U-tube indicating vaccum	Closed U-tube indicating absolute pressure

If P is atmospheric pressure, this last one is a barometer.

These are called U-tube manometers.



2. A turbine is supplied with steam at a gauge pressure of 1.4 MPa. After expansion in the turbine the steam flows into a condenser which is maintained at a vaccum of 710 mm Hg. The

barometric pressure is 772 mm Hg. Express the inlet and exhaust steam pressure in pascals (absolute). Take the density of mercury as 13.6×10^3 kg/m³

The atmospheric pressure is

$P_o = \rho g z_o = 13.6 \times 10^3 \text{ kg/m}^3 \times 9.81 \text{ m/s}^2 \times 0.772 \text{ m}$

$= 1.03 \times 10^5$ Pa

Inlet steam pressure is, $= 1.4$ MPa ← gauge pressure

$= \left((1.4 \times 10^6) + (1.03 \times 10^5)\right)$ Pa ← absolute pressure

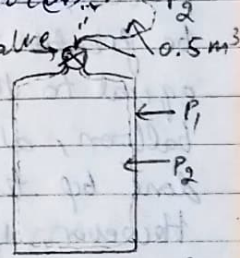
$= 15.03 \times 10^5$ Pa $= 1.503$ MPa

Condenser pressure is, $= 710$ mm ← Vacuum pressure

$= (0.772 - 0.710)\text{m} \times 9.81 \text{ m/s}^2 \times 13.6 \times 10^3 \text{ kg/m}^3$ ← absolute pressure

$= 6.827 \times 10^3$ Pa $= 8.27$ kPa

3. Gas from a bottle of compressed helium is used to inflate an inelastic flexible balloon, originally folded completely flat to a volume of 0.5 m³. If the barometer reads 760 mm Hg, what is the amount of work done upon the atmosphere by the balloon? Sketch the system before and after the process.



The firm line P_1 shows the boundary of the system before the process and the dotted line P_2 shows the boundary after the process. The displaced work is,

$W_2 = \int_{\text{ballon}} P dV + \int_{\text{bottle}} P dV$

$= P \Delta V + 0$

$= 101.325 \times 10^3 \text{ N/m}^2 \times 0.5 \text{ m}^3 = 50.66$ kJ

This is positive because work is done by the system. Work done by the atmosphere is -50.66 kJ. Since the wall of the bottle is rigid, there is no pdV work involved in it.

It is assumed the pressure in the balloon is same as atmosphere at all times, since the balloon fabric is light, inelastic

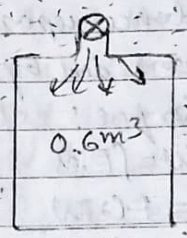
and unstressed. If the balloon were elastic and stressed during the filling process, the work done by the gas would be greater than 50.66 kJ by an amount equal to the work done in stretching the balloon, although the displacement work done by the atmosphere is still −50.66 kJ. However, if the system includes both the gas and the balloon, the displacement work would be 50.66 kJ as estimated above.

4. When the valve of the evacuated bottle is opened, atmospheric air rushes into it. If the atmospheric pressure is 101.325 kPa and 0.6 m^3 of air (measured at atmospheric conditions) enters into the bottle, calculate the work done by air.

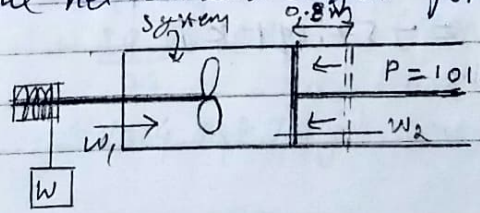
The displacement work done by air is,

$$W_d = p\Delta V = 101.325\ kN m^{-2} \times 0.6\ m^3$$
$$= 60.8\ kJ$$

$P_{atm} = 101.325$ kPa



5. A piston and cylinder machine containing a fluid system has a stirrer device in the cylinder. The piston is frictionless and it is held down against the fluid due to the atmospheric pressure of 101.32k kPa. The stirring device is turned 10,000 revolutions with an average torque against the fluid of 1.275 kNm. Meanwhile the piston of 0.6 m diameter moves out 0.8 m. Find the net work transfer for the system.



a) Work done by the stirring device upon the system is.

$W_1 = 2\pi\tau n$

$\tau = 1.275\ Nm$

$n = 10000$

$\therefore W_1 = 2 \times 3.142 \times 1.275\ Nm \times 10000$

$= 80.088\ kJ$ (-ve work for the system)

work done by torque:

$W = \int \vec{F} \cdot d\vec{s}$

$= F(2\pi s)$

$= 2\pi F s = 2\pi\tau$

$\because Fs = \tau$ (torque)

b) Work done by the system upon the surrounding due to expansion.

$W_2 = \int P dV = P\Delta V = P(\Delta L)A$

$= 101.325\ kN m^{-2} \times 0.8\ m \times \pi \times (0.3)^2 m^2$ $\left(\because r = \frac{d}{2} = \frac{0.6m}{2} = 0.3m\right)$

$= 22.919\ kJ$

(+ve work by the system)

Hence the net work transfer for the system is

$W = W_1 + W_2 = -80.088\ kJ + 22.919\ kJ$

$= -57.169\ kJ$

6. If it is desired to melt aluminium with solid state specific heat $0.9\ kJ\ kg^{-1}K^{-1}$, latent heat $390\ kJ\ kg^{-1}$, atomic weight $27\ g\ mol^{-1}$, density in molten state $2400\ kg\ m^{-3}$ and final temperature 700°C, find out how much metal can be melted per hour with the above kW rating of 2.17×10^3 kW. Other data are; a) The melting point of aluminium is 660°C. b) Efficiency of the furnace is 70%. c) Initial temp 15°C. d) Volume of furnace $2.8\ m^3$. Also find the mass of aluminium that the above furnace will hold. Specific heat in molten state = $29.93\ J\ K^{-1}\ mol^{-1}$ / atomic weight = $\frac{29.93}{27}\ kJ\ kg^{-1}\ K^{-1}$

Heat required per kg of aluminum = Heat required to raise the temperature from 15°C to melting point + Latent heat + Heat required to raise the temp of molten Al from 660°C to 700°C

$= 0.9\ kJ\ kg^{-1}K^{-1} \times (660-15)K + 390\ kJ\ kg^{-1} + \frac{29.93}{27}\ kJ\ kg^{-1}K^{-1}(700-660)K$

$= 1014.84\ kJ\ kg^{-1}$

For 70% efficient furnace, heat to be supplied is,

$= \frac{1014.84}{0.7} = 1449.77 \text{ kJ/kg}$

Given power $= 2.17 \times 10^3 \text{ kW} = 2.17 \times 10^3 \text{ kJ s}^{-1}$

$= \frac{2.17 \times 10^3}{\frac{1}{3600}} \text{ kJ h}^{-1} = 2.17 \times 10^3 \times 3600 \text{ kJ h}^{-1}$

With the given power, the rate at which aluminium can be melted is,

$= \frac{2.17 \times 10^3 \times 3600 \text{ kJ h}^{-1}}{1449.77 \text{ kJ/kg}}$

$= 5388.43 \text{ kg h}^{-1} = 5.38843 \text{ tonnes/h}$

Mass of aluminium that can be held $= 2.18 \text{ m}^3 \times 2400 \text{ kg m}^{-3}$

$= 5.23 \text{ tonnes}$.

Only 70% of this supplied power will be utilized to melt Al. Therefore power available for melting is,

$= 2.17 \times 10^3 \times 3600 \times 0.7 \text{ kJ h}^{-1}$

$= 5468.4 \times 10^3 \text{ kJ h}^{-1}$

With this available power. Al that could be melted per hour is,

$= \frac{\text{Power available}}{\text{Energy required per kg}}$

$= \frac{5468.4 \times 10^3 \text{ kJ h}^{-1}}{1014.84 \text{ kJ kg}^{-1}}$

$= 5.3884 \text{ kg h}^{-1} \times 10^3$

$= 5.3884 \text{ tonnes h}^{-1}$

Mass of Al that can be held in molten

$=$ volume of furnace $\times$ density of Al

$= 2.18 \text{ m}^3 \times 2400 \text{ kg m}^{-3} = 5.232 \times 10^3 \text{ kg}$

7. A stationary mass of gas is compressed without friction from an initial state of 0.3 m^3 and 0.105 MPa to a final state of 0.15 m^3 and 0.105 MPa, the pressure remaining constant during the process. There is a transfer of 37.6 kJ of heat from the gas during the process. How much does the internal energy of the gas change?

First law for closed system gives,

$$Q = \Delta U + W$$

$$\therefore Q_{12} = (U_2 - U_1) + W_{12}$$

Here, $W_{12} = \int_{V_1}^{V_2} p\,dV = p(V_2 - V_1)$

$$= 0.105 \times 10^6 \, N m^{-2} \times (0.15 - 0.3) m^3$$

$$= -15.75 \, kJ$$

Heat transfer from the gas is,

$$Q_{12} = -37.6 \, kJ \quad (-ve)$$

$$\therefore \Delta U = Q_{12} - W_{12}$$

$$= -37.6 kJ - (-15.75 kJ)$$

$$= -21.85 \, kJ$$

The internal energy of the gas decreases by 21.85 kJ in the process.

8. When a system is taken from state a to state b, in the figure, along path acb, 84 kJ of heat flow into the system, and the system does 32 kJ of work.

a). How much will the heat that flows into the system along path adb be, if the work done is 10.5 kJ?

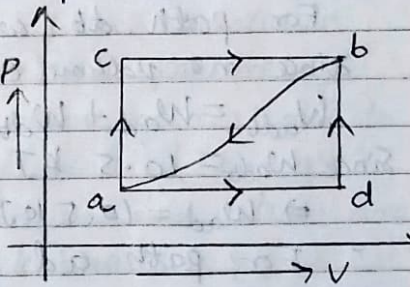
b). When the system is returned from b to a along the curved path, the work done on the system is 21 kJ. Does the system absorb or liberate heat, and how much of the heat is absorbed or liberated?

c). If $U_a = 0$ and $U_d = 42 kJ$, find the heat absorbed in the process ad and db.

Given:

$Q_{acb} = 84 \, kJ$

$W_{acb} = 32 \, kJ$



We have from first law,

$$Q_{acb} = (U_b - U_a) + W_{acb}$$

Heat Q and work W are path functions but internal energy is a point function.

$$\therefore U_b - U_a = Q_{acb} - W_{acb}$$

$$= (84 - 32) kJ = 52 \, kJ$$

a) For path adb first law says,

$$Q_{adb} = (U_b - U_a) + W_{adb}$$

Given $W_{adb} = 10.5\ kJ$

$\therefore Q_{adb} = 52\ kJ + 10.5\ kJ$

$= 62.5\ kJ$

b) For return path ba

$Q_{ba} = (U_a - U_b) + W_{ba}$

$= -52\ kJ - 21\ kJ$

$= -73\ kJ$

*Work done on the system -Ve
*Initial internal energy is U_b and final is U_a

The system liberates 73 kJ of heat

c) For path ad work is W_{ad}

For path db work is $W_{db} = 0$ since no volume change involved.

$W_{adb} = W_{ad} + W_{db} = W_{ad} + 0$

Since $W_{adb} = 10.5\ kJ$ (given)

$\Rightarrow W_{ad} = 10.5\ kJ$

For path ad with $U_a = 0$ and $U_d = 42\ kJ$ (given), from first law

$Q_{ad} = U_d - U_a + W_{ad}$

$= (42 - 0)\ kJ + 10.5\ kJ$

$= 52.5\ kJ$

Since $Q_{adb} = 62.5\ kJ$ (calculated in (a)) and $Q_{adb} = Q_{ad} + Q_{db}$

$\Rightarrow 62.5\ kJ = 52.5\ kJ + Q_{db}$

$\Rightarrow Q_{db} = 62.5\ kJ - 52.5\ kJ = 10\ kJ$

9. The internal energy of a certain substance is given by the following equation

$$u = 3.56\,pv + 84$$

where u is given in kJ/kg, p is in kPa and v is in m^3/kg. A system composed of 3 kg of this substance expands from an initial pressure of 500 kPa and a volume of $0.22\ m^3$ to a final pressure of 100 kPa in a process in which pressure and volume are related by $pv^{1.2} = \text{constant}$.

a) If the expansion is quasi-static, find Q, ΔU and W for the process.

b) In another process the same system expands according to the same pressure-volume relationship as in part (a), and from the same initial

state to the same final state as in part (a), but the heat transfer in this case is 30 kJ. Find the work transfer for this process.

c) Explain the difference in work in parts (a) and (b).

(a) $u = 3.56\,pv + 84$

$\Delta u = u_2 - u_1 = 3.56\,(P_2 v_2 - P_1 v_1)$

and $\Delta U = m\,\Delta u$

$= 3.56\,(P_2 m v_2 - P_1 m v_1)$

$= 3.56\,(P_2 V_2 - P_1 V_1)$

where $mv_1 = V_1$ and $mv_2 = V_2$

with $pv^{1.2} = \text{constant}$

$\Rightarrow P_1 v_1^{1.2} = P_2 v_2^{1.2}$

$\Rightarrow P_1 V_1^{1.2} = P_2 V_2^{1.2}$

$V_2 = V_1 \left(\frac{P_1}{P_2}\right)^{1/1.2} = 0.22\,m^3 \left(\frac{500\,kPa}{100\,kPa}\right)^{\frac{1}{1.2}}$

$= 0.22 \times 5^{1/1.2}\,m^3$

$= 0.8412\,m^3$

$\therefore \Delta U = 3.56\,(P_2 V_2 - P_1 V_1)$

$= 3.56\,(100\,kPa \times 0.8412\,m^3 - 500\,kPa \times 0.22\,m^3)$

$= -92.1328\,kJ$ // $[\because 1\,kPa\,m^3 = 1\,kJ]$

Work done for a quasi-static process is, with $PV^{1.2} = K$ (constant)

$W = \int P\,dV = \int_{V_1}^{V_2} \frac{K\,dV}{V^{1.2}} = \left[\frac{K V^{1-1.2}}{1-1.2}\right]_{V_1}^{V_2}$

$= K \left[\frac{V^{-0.2}}{-0.2}\right]_{V_1}^{V_2} = K\left[\frac{V_1^{-0.2}}{0.2} - \frac{V_2^{-0.2}}{0.2}\right]$

$= \frac{K}{0.2}\left[V_1^{-0.2} - V_2^{-0.2}\right]$

with $P_1 V_1^{1.2} = P_2 V_2^{1.2} = K$

$W = \frac{1}{0.2}\left[P_1 V_1^{1.2} \cdot V_1^{-0.2} - P_2 V_2^{1.2} V_2^{-0.2}\right]$

$= \frac{1}{0.2}\left(P_1 V_1 - P_2 V_2\right]$

$= \frac{1}{0.2}\left[500\,kPa \times 0.22\,m^3 - 100\,kPa \times 0.8412\,m^3\right]$

$= 129.4\,kJ$ //

Then from first law of thermodynamics,

$Q = \Delta U + W$

$= -92.1328\,kJ + 129.4\,kJ$

$= 37.2672\,kJ$ //

(b) Here $Q = 30\,kJ$.
Since the end states are the same, ΔV would remain the same as in (a).
$\therefore W = Q - \Delta V$
$= 30\,kJ - (-92.1328)\,kJ$
$= 122.1328\,kJ$

(c) Since work transfer is different for some PV relation then for part (b) the work will not be equal to $\int P\,dV$. Hence the process (b) is not quasi-static.

Note: Only for quasi-static process $W = \int P\,dV$.

10. A piston cylinder device operates 1kg of fluid at 20atm pressure with initial volume $0.04\,m^3$. Fluid is allowed to expand reversibly following $PV^{1.45} = C$ so that the volume becomes double. The fluid is cooled at constant pressure until the piston comes back. Determine the work done in each process.

Given: $m = 1\,kg$

$P_i = 20\,atm = 20 \times 101.325\,kPa$

$V_i = 0.04\,m^3$

$V_f = 2 \times 0.04\,m^3$

$P_f = \dfrac{C}{V_f^{1.45}}$ where $C = P_i V_i^{1.45}$

$= P_i \left(\dfrac{V_i}{V_f}\right)^{1.45} = 20 \times 101.325\,kPa \left(\dfrac{1}{2}\right)^{1.45}$

First is an expansion of the form $PV^{1.45} = C$ or $PV^{\gamma} = C$ form for which work done is

$$W = \int_{V_i}^{V_f} P\,dV = \frac{P_f V_f - P_i V_i}{1-\gamma}$$

$$= \frac{[20 \times 101.325\,kPa \times (0.5)^{1.45} \times 2 \times 0.04\,m^3 - 20 \times 101.325\,kPa \times 0.04\,m^3]}{1 - 1.45}$$

$$= \frac{20 \times 101.325 \times 0.04\,[2 \times (0.5)^{1.45} - 1]\,kJ}{-0.45}$$

$= 48.268\,kJ$

Second it in isobaric compression for which work done in,

$$W = P\int dV = P\Delta V = P(V_f - V_i)$$
$$= P(0.04 - 2 \times 0.04)\,m^3$$
$$= -0.04(P)\,m^3$$

Substitute value of final pressure P_f for P.

11. When a stationary mass of gas was compressed without friction at constant pressure, its initial state of $0.4\ m^3$ and 0.105 MPa was found to change to final state of $0.20\ m^3$ and 0.105 MPa. There was a transfer of 42.5 kJ of heat from the gas during the process. Determine ~~those~~ the change in internal energy of the gas?

From first law of Thermodynamics,

$$Q = W + \Delta V$$
$$\therefore \Delta V = Q - W$$

Q is given and it is -ve, it is because heat leaving from the system is -ve,

$$\therefore Q = -42.5\ kJ$$

and $W = \int P dV = P\int_{V_i}^{V_f} dV$ (for constant pressure)

$$= 0.105\ MPa \times (V_f - V_i)$$
$$= 0.105\ MPa\,(0.20 - 0.40)\,m^3$$
$$= 0.105 \times (-0.2) \times 10^6\ J$$
$$= -0.021 \times 10^3\ kJ$$
$$= -21\ kJ$$

$$\therefore \Delta V = Q - W = -42.5\ kJ - (-21\ kJ)$$
$$= \underline{-21.5\ kJ}$$

12) 0.44 kg of air at 1800°C expands adiabatically to 3 times its original volume and during the process there is a fall in temperature to 150°C. The work done during the process is 52.5 kJ. Calculate C_p and C_v.

$m = 0.44$ kg

$T_i = 1800°C \rightarrow (1800 + 273.15)K = 2073.15$

$V_i = V$

$T_f = 150°C \rightarrow (150 + 273.15)K = 423.15K$

$V_f = 3V$

$W = 52.5$ kJ

For adiabatic expansion the equation of state is,

$$PV^{\gamma} = \text{constant}$$

or $P_i V_i^{\gamma} = P_f V_f^{\gamma}$

or $\frac{P_i}{P_f} = \left(\frac{V_f}{V_i}\right)^{\gamma} \longrightarrow (a)$

From ideal gas equation,

$P_i V_i = nR_0 T_i$ } where R_0 is
$P_f V_f = nR_0 T_f$ } universal gas constant

Dividing these.

$$\frac{P_i}{P_f} = \frac{V_f}{V_i}\frac{T_i}{T_f} \longrightarrow (b)$$

From (a) and (b), diminate P_i/P_f,

$$\Rightarrow \left(\frac{V_f}{V_i}\right)^{\gamma} = \frac{V_f}{V_i}\frac{T_i}{T_f}$$

$$\Rightarrow \left(\frac{V_f}{V_i}\right)^{\gamma-1} = \frac{T_i}{T_f}$$

$$\Rightarrow \gamma - 1 = \frac{\ln(T_i/T_f)}{\ln(V_f/V_i)} = \frac{\ln\left(\frac{1800}{+273.15}/\frac{273.15}{150+6}\right)}{\ln(3V/V)}$$

$$= \frac{\ln(4.888)}{\ln(3)} = 1.4464$$

$$\Rightarrow \gamma = 1.4464 + 1 = 2.4464$$

and $\gamma = \frac{C_p}{C_v}$. We are yet to find C_p and C_v separately.

Work done in such adiabatic process is,

$$W = \frac{1}{1-\gamma}(P_f V_f - P_i V_i)$$

With $PV = mRT$ we can have $P_i V_i = mRT_i$ and $P_f V_f = mRT_f$.

Substituting these two in above work done expression,

$$W = \frac{1}{1-\gamma}[mRT_f - mRT_i] = \frac{mR}{1-\gamma}[T_f - T_i]$$

Here R is the characteristic gas constant and m is the mass of the gas.

$$\Rightarrow 52.5\,kJ = \frac{0.44\,kg\,R\,(150^{\circ}C - 1800^{\circ}C)}{(1-2.4464)}$$

$$\Rightarrow R = \frac{(1-2.4464)\times 52.5\,kJ}{0.44\,kg \times (-1650\,K)}$$

$$= 0.1046\,kJ\,kg^{-1}K^{-1}$$

and $C_p - C_v = R$.

We have now calculated γ and R.
They relates C_p and C_v by,

$$\frac{C_p}{C_v} = \gamma \quad \Rightarrow C_p = \gamma C_v$$

and $C_p - C_v = R$

$$\Rightarrow \gamma C_v - C_v = R$$

$$\Rightarrow C_v = \frac{R}{\gamma - 1} = \frac{0.1046 \text{ kJ kg}^{-1}\text{K}^{-1}}{1.4464}$$

$$= 72.314 \text{ J kg}^{-1}\text{K}^{-1}$$

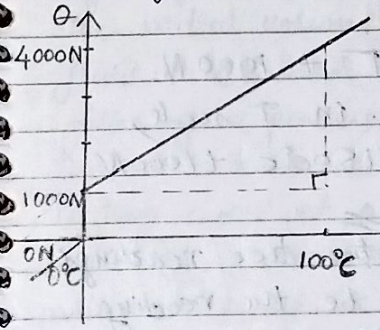
With this,

$$C_p = R + C_v$$

$$= 104.6 \text{ J kg}^{-1}\text{K}^{-1} + 72.314 \text{ J kg}^{-1}\text{K}^{-1}$$

$$= 176.914 \text{ J kg}^{-1}\text{K}^{-1}.$$

13. A new scale N of temperature is divided in such a way that the freezing point of ice is 1000 N and the boiling point is 4000 N. What is the temperature reading on this new scale when the temperature is 1500 °C? At what temperature both the celsius and the new scale reading would be the same?



This problem can be illustrated and solved by using the graph besides. Let the plot be with T axis in °C and θ axis in N (new temperature scale unit).

For freezing point: $T = 0$ °C, $\theta = 1000$ N
For boiling point: $T = 100$°C, $\theta = 4000$ N

With these two points the graph is plotted with a straight line assuming that the scales are linear and hence their relation also. The above graph represents the relation between the two scales.

For straight line the relation is

$y = mx + c$, m = slope, c = intercept

Here it is, $\theta = mT + c$

where $m = \frac{4000\text{N} - 1000\text{N}}{100°\text{C} - 0°\text{C}} = \frac{3000\text{N}}{100°\text{C}}$

$= 30 \text{ N/°C}$

and the intercept $C = 1000\,N$.
Therefore the relation between the two units is,

$$\theta = 30\,N/^{\circ}C\,T + 1000\,N$$

Now for 1500°C. in T scale,

$$\theta = 30\,N/^{\circ}C \times 1500^{\circ}C + 1000\,N$$
$$= 46000\,N$$

To know when the two readings be same, let R be the reading which is same for both scales. They at that temperature, by the relation,

$$R(in\,N) = 30\,N/^{\circ}C \times R(in\,^{\circ}C) + 1000\,N$$
$$\Rightarrow R(in\,N) = 30R(in\,N) + 1000\,N$$
$$\Rightarrow -29R(in\,N) = 1000\,N$$
$$\Rightarrow R \times (-29\,N) = 1000\,N$$

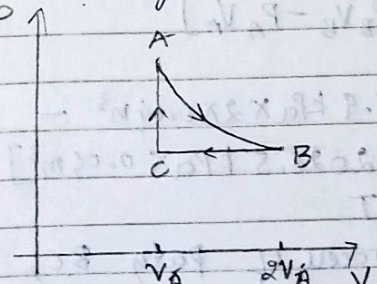
$\theta = RN$ and $T = R^{\circ}C$
(R: magnitude, N: unit; R: magnitude, °C: unit)

Substituting in $\theta = 30\,N/^{\circ}C\,T + 1000\,N$
we get, $RN = 30\,N/^{\circ}C \times R\,^{\circ}C + 1000\,N$

$$\Rightarrow RN = 30RN + 1000\,N$$
$$\Rightarrow -29RN = 1000\,N$$
$$\Rightarrow R = -1000/29 = -34.483$$

Hence at $-34.483^{\circ}C$ both scales read same.

14. A piston-cylinder device operates 1 kg of a fluid at 20 atm. of pressure. The initial volume is $0.04\,m^3$. The fluid is allowed to expand reversibly following a process $PV^{1.4}$ = constant so that the volume becomes double. The fluid is then cooled at a constant pressure until the piston comes back to the original position. Keeping the piston unaltered, heat is added reversibly to restore it to the initial pressure. Calculate the work done in the cycle.



As per the description of the problem this is the PV diagram. Path AB is adiabatic, path BC is isobaric and path CA is isochoric. Following are the given quantities in the problem:

$m = 1\,kg$

$V_A = V = 0.04\,m^3$

$V_B = 2V = 2 \times 0.04\,m^3$

$P_A = 20\,atm = 20 \times 101.325\,kPa = 2026.5\,kPa$

$\gamma = 1.4$

$PV^{1.4} = \text{constant}$ for path AB.

$\therefore P_A V_A^{1.4} = P_B V_B^{1.4}$

$P_B = P_A \left(\frac{V_A}{V_B}\right)^{1.4} = 20\,atm \left(\frac{V}{2V}\right)^{1.4}$

$= \frac{20\,atm}{2^{1.4}} = 7.5786\,atm = 7.5786 \times 101.325\,kPa$

$= 767.8999\,kPa$

For adiabatic process of path AB,

$W_{AB} = \frac{1}{1-\gamma}[P_B V_B - P_A V_A]$

$= \frac{1}{1-1.4}[767.9\,kPa \times 2 \times 0.04\,m^3 - 2026.5\,kPa \times 0.04\,m^3]$

$= 49.07\,kJ$

For isobaric process of path BC,

$W_{BC} = P \int dV = P\Delta V = P(V_{final} - V_{ini})$

$= P_B(V_C - V_B)$

Here, $V_B = 2 \times 0.04\,m^3$

$V_C = V_A = 0.04\,m^3$

$\therefore W_{BC} = 767.9\,kPa \times (0.04\,m^3 - 2 \times 0.04\,m^3)$

$= -30.716\,kJ$

For isochoric process of path CA since there will be no change in the volume, work done will be 0.

$\therefore W_{CA} = 0\,kJ$

Hence work done in the cycle is,

$W = W_{AB} + W_{BC} + W_{CA}$

$= 49.07\,kJ - 30.716\,kJ + 0\,kJ$

$= \underline{18.354\,kJ}$